

10 Inheritance Problems (Java)

1 Shape & Circle/Rectangle

Create a superclass `Shape` with a method `area()`. Create subclasses `Circle` and `Rectangle` that calculate the area using overridden `area()` methods.

2 Vehicle & Car/Bike

Create a superclass `Vehicle` with `start()` method. Create subclasses `Car` and `Bike` that override `start()` to print "Car started" and "Bike started".

3 Employee & Manager/Developer

Create a superclass `Employee` with fields `name` and `salary`, and a method `display()`. Subclasses `Manager` and `Developer` add extra fields (`bonus` or `project`) and override `display()`.

4 Bank Account & Savings/Current

Superclass `BankAccount` with methods `deposit()` and `withdraw()`. Subclasses `SavingsAccount` and `CurrentAccount` override `withdraw` rules (e.g., minimum balance).

5 Animal & Dog/Cat

Superclass `Animal` has method `makeSound()`. Subclasses `Dog` and `Cat` override it to print "Bark" and "Meow". Demonstrate dynamic method dispatch.

6 Person & Student/Teacher

Superclass `Person` with `name` and `age`. Subclasses `Student` and `Teacher` add `grade` or `subject`. Override `displayInfo()` to print all info.

7 Library Book & EBook/PrintedBook

Superclass `Book` with title and author. Subclasses `EBook` and `PrintedBook` add `fileSize` or `pages`. Override `displayInfo()` to include subclass info.

8 Employee & FullTime/PartTime

Superclass `Employee` with `calculateSalary()`. Subclasses `FullTimeEmployee` and `PartTimeEmployee` override `calculateSalary()` to calculate based on monthly salary or hourly rate.

9 Shape & 2D/3D

Superclass `Shape` with method `area()`. Subclasses `TwoDShape` and `ThreeDShape`. Override `area()` for 2D shapes and `volume()` for 3D shapes.

10 Device & Phone/Laptop

Superclass `Device` with method `switchOn()`. Subclasses `Phone` and `Laptop` override `switchOn()` to print device-specific messages.